

A Study on Affordable Manipulation in Virtual Reality Simulations: Hand-Tracking versus Controller-Based Interaction

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ABSTRACT

When it comes to Virtual Reality (VR) simulations, controllers are used to interact with the virtual world. However, controllers are an approximation, and a possible alternative is represented by optical hand-tracking. The goal of the present paper is to compare the two solutions to investigate the best way to implement interactions in VR. Results showed that, despite its limitations, hand-tracking enables realistic interactions that are more consistent with the real world.

Index Terms: Human-machine interaction—Virtual Reality—Hand-tracking—Simulation

1 INTRODUCTION

Immersive Virtual Reality (VR) technologies are widely adopted in several contexts, such as entertainment [13], simulation [12] and education [7]. When VR is used in educational context, as an alternative or as a complementary module to existing courses, it is often adopted for teaching procedural tasks where the user is required to interact with objects and follow a series of actions to achieve a particular goal. This is due often to the capability of VR to simulate realistic situations and replicate tasks with a high level of detail, thus offering a realistic experience that can be compared with real-world scenarios.

When it comes to interaction with virtual objects, VR simulations offer different Human-Machine Interaction (HMI) methods that can be considered depending on the nature of the tasks requested by the virtual experience. Usually, users are provided with one or more controllers that are tracked and offer a series of input buttons; this solution is inexpensive and scalable, and allow the users to interact with a variety of objects, but it is often an abstraction of the real interactions and its adoption can reduce the realism and the immersion of the experience.

When the simulated procedure requires the use of tools that need to be handled with particular care and must be used in a specific way (i.e., a firefighting shovel that must be held with a specific orientation to avoid risks), a possible solution is to use real objects that can be tracked in real-time. However, this solution is characterized by a series of limitations; on the one hand, it can be adopted only when the considered object is big enough to be tracked and the relative weight of the tracker can be considered negligible. On the other hand, it is feasible only for procedures where the use of the considered tool is predominant w.r.t. other interactions, otherwise it would be

necessary to use a large number of trackers or implement different types of interactions in the same application, possibly resulting in a simulation undermined by inconsistencies that could hinder the overall quality of the experience.

A possible alternative is the use of gloves that track the users' fingers and possibly offer haptic feedback, but they are usually very expensive or present technical limitations. Finally, another option is to rely on the optical hand-tracking technology offered by the Head-Mounted Displays (HMDs), that despite its limitations (e.g., the impossibility to track the users' hands when they are outside the field of view of the HMD) is a more affordable and scalable solution that can offer natural interactions without breaking the immersion of the experience.

Starting from these considerations, this paper aims to compare two different, affordable interaction methods (controller-based and hand-tracking) for interacting with virtual environments and perform different tasks, such as grabbing objects, remove clothes or covers, activate tools, etc. The comparison was conducted in the context of a CBRN procedure (Contamination Control Area & Collective Protection, CCA COLPRO), that was chosen as a case study due to the wide scope of manual actions that must be performed in order to successfully complete the considered task. The two methods were compared through a within-subjects user study that involved 15 participants and evaluated a series of subjective metrics such as the usability, the attractiveness of the experience, the overall immersion, etc.

2 RELATED WORKS

Nowadays, VR technologies are often adopted in training contexts due to their effectiveness in transferring procedural knowledge over other type of learning materials [3]; this is true, in particular, when the considered procedure relies heavily on the manipulation and the movement of objects in the virtual environment and it is required to learn a series of hands-on actions to successfully complete the considered task [4]. It is therefore necessary to focus on the best way to implement interactions in a VR simulation, with a particular focus on realism and naturalness.

Looking at the literature, it is possible to find a series of studies that investigated how to interact with virtual objects. Controller-based applications are a widespread solution [4,12], since controllers are included with HMDs and their cost is usually low [18]. However, controllers must be continuously held in hand, and this could represent a negative aspect in scenario where bare-hands interactions (such as grabbing, pushing, pulling, etc.) are required [1]. In similar situations, in fact, it is not uncommon that users ask for alternative interfaces, citing the need for more-natural interaction methods [12].

One of the alternatives is represented by haptic gloves, wearable devices that tracks the positions of the fingers using various technologies. However, as shown in [14], less expensive solutions based on these devices are usually affected by technical limitations (such as poor tracking performance), whereas newer and more reliable products are often costly and are not included in consumer HMDs. A possible affordable alternative can be found in the optical-based

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hand-tracking that is already integrated in consumer HMDs (e.g., Meta Quest 1 and 2). This solution is characterised by a lack of passive feedback that puts it behind haptic gloves in terms of realism [6], but it can offer natural interactions that are normally impossible with hand controllers.

Looking at the literature, it is possible to find a number of works researching several aspects of this technology, but there is a lack of analysis and comparison between hand-tracking and controller-based experiences with a focus on the naturalness and the realism of the interactions. The study presented in [5], for example, investigated the best gestures to implement interactions in a virtual environment, resulting in a ranking of gestures for each action that was considered by the authors, without studying the advantages associated with the use of hand-tracking for those particular actions. Moreover, in [19] the authors compared the use of hand-tracking and controllers (with a focus on joystick controls), without focusing the analysis on the naturalness and the realism of the interactions but considering other aspects of usability.

3 METHODOLOGY

This paper presents a comparison between two HMI methods. To perform the comparison, two versions of the same application were used; the two versions are identical except for the interaction method: one leverage the controllers, whereas the other one uses the hand-tracking technology of the HMD (i.e., a Meta Quest 2).

3.1 Case Study

The considered case study is a CBRN procedure in which the operators are required to decontaminate and clean a potentially contaminated person. During the procedure, operators perform several manual tasks, such as:

- checking for contamination using a probe;
- decontaminating the person using a liquid dispenser;
- removing different pieces of clothing, such as over-boots, uniform (first by removing the hood, then by opening a zip), gloves, and the mask (with the corresponding filters);

3.2 Software

The two versions of the application were developed for immersive VR HMDs and, in particular, were targeted towards the Meta Quest 2. The controller-based version used the Meta Quest 2 controllers and the corresponding inputs buttons to interact with the environment (Fig. 1a), whereas the hand-tracking one leveraged the Meta Quest 2 built-in hand-tracking technology that tracks the users' hands via the HMDs cameras (Fig. 1b). Both versions were developed in the Unity game engine, using the SteamVR framework to handle the controllers and the Mixed Reality Toolkit (MRTK), an open source suite provided by Microsoft, to manage the hand-tracking logic. These two solutions were the state of the art at the time of the development and were preferred over custom implementations. The 3D assets were created using Blender, together with image-editing software for the creation of the textures. The logic and algorithms controlling the virtual 3D objects were written in the C# programming language.

In both versions of the application, the user can navigate freely the virtual environment (by walking in the real world, or teleporting inside the application) and interact with the 3D objects to complete a series of tasks. In the controller-based version, the interactions are mediated by the controllers and the corresponding input buttons; to interact with an object, the user must touch it with his or her virtual hand (that is a 3D object corresponding to the real-time position of the controller) and press a button to start the interaction. If the user want to grab an object, he or she must keep the button pressed and release it to drop the object. If the user want to use

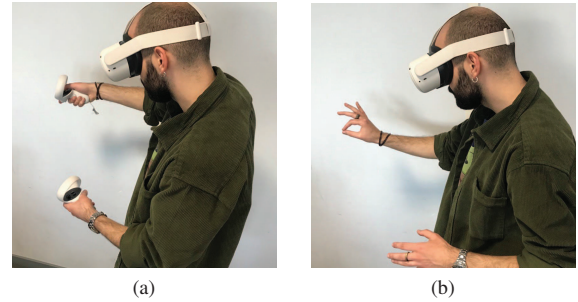


Figure 1: User interacting with the application in the controller-based version (a) and in the hand-tracking version (b).

an object (for example, eject the liquid contained in a dispenser) he or she can press another button to activate it. In this version, the hand models can assume a set of limited poses, determined in the implementation phase (Fig 2a). In the hand-tracking version of the application, the interaction are mapped directly onto the hands of the user; he or she must keep his or her hands in front of the HMD, and perform different gestures. To interact with an object the user must perform a "Pinch" gesture, whereas in case of grabbing he or she must perform the same gesture and hold it until the end, simulating the natural interaction for grabbing small objects. To use an object (e.g., the dispenser) the user must perform a "Fist" gesture, opening and closing the thumb to activate it (e.g., eject the liquid), thus approximating the real interaction. In this version, the hand model are animated in real-time and follow the poses of the user's hands (Fig. 2b).

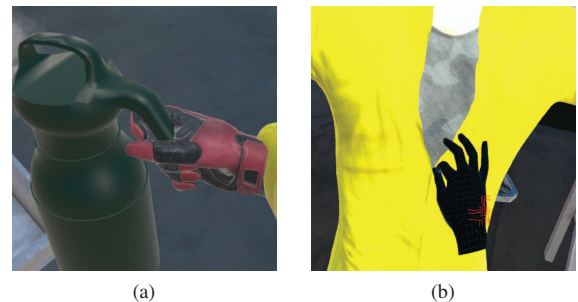


Figure 2: The virtual hand assume a pre-determined pose in the controller-based version (a), and it is animated in real time in the hand-tracking version (b).

4 EXPERIMENT

This section describes the user study conducted to compare the two HMI methods. A within-subject study was carried out involving 15 users (4 males and 11 females) aged between 23 and 51 ($\bar{x} = 32.20$, $sd = 9.09$). The within-subject approach was chosen to generalize the results and avoid that the order in which the HMI methods were experienced could introduce a bias. The participants were recruited from the staff of the authors' affiliations.

4.1 Procedure

The structure of the experimental procedure is resumed in Fig. 3. First, the participants were asked to fill a brief pre-test questionnaire aimed at collecting general information. The participants were randomly subdivided into two groups: a first one called Controller-first

(C in the following, seven participants) and a second one called Hand-tracking-first (H, eight participants). Afterward, participants from group C experienced the controller-based version of the application, whereas the ones from group H experienced the hand-tracking version. All the participants were then asked to fill a mid-test questionnaire to evaluate the experience up to that moment. Then, the participants from group C experienced the hand-tracking version of the application, whereas the ones from group H experienced the controller-based version. Finally, a post-test questionnaire was administered to all participants to evaluate the second HMI method that they experienced and compare the two methods.

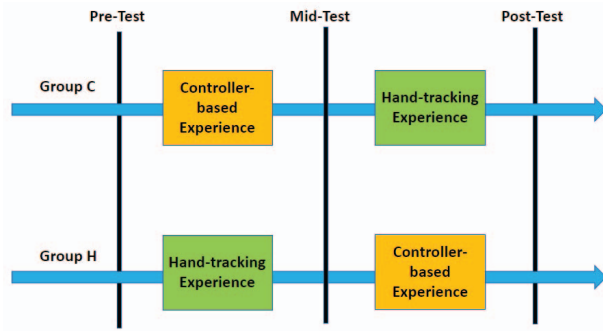


Figure 3: Organization of the testing procedure for the two groups (C and H).

4.2 Questionnaires

The questionnaires (pre-, mid-, and post-test) were aimed to evaluate the immersive VR experience, with a focus on the devised HMI method.

In particular, the pre-test questionnaire encompassed general questions on the participant, investigating aspects such as age, gender and previous experience with VR technologies, with a focus on the use of controllers or hand tracking technologies.

The mid-test questionnaire encompassed items aimed at collecting subjective measurements on the experience and the HMI method. These items are subdivided in the following sections:

- the first section evaluated the general usability of the considered system (with a focus on the HMI method) and was based on the System Usability Scale (SUS) [2]. It consisted of 10 statements to be scored on a 1-to-5 Likert scale.
- The second section evaluated the overall presence associated with the considered system (with a focus on the HMI method), thus, in particular, the naturalness and realism of the interactions. It was based on a questionnaire from Sutcliffe et al. [15] and it consisted of five statements to be scored on a 1-to-5 Likert scale.
- The third section evaluated the perceived enjoyment associated with the considered system (with a focus on the HMI method) and was based on a questionnaire from Toker and Isler [16]; it consisted of five statements to be scored on a 1-to-5 Likert scale.
- The fourth section consisted of a subset of the AttrakDiff questionnaire [9] evaluating the Attractiveness (ATT) and Hedonic Quality Stimulation (HQ-S) of the experience [10]; it consisted of 14 pairs of terms to which each participant was asked to assign a value on a 1-to-7 scale (with low values corresponding to a positive evaluation).

- The fifth section investigated in depth various usability factors (including the fidelity of the simulation to real life experiences, the feeling of presence and the overall satisfaction) and was based on the VRUSE questionnaire [11]; it consisted of five statements to be scored on a 1-to-5 Likert scale.
- The sixth section investigated the mental workload and was based on the SIM TLX questionnaire [8], and it consisted of 10 statements to be scored on a 1-to-10 scale.
- An additional, non-numbered section was aimed to collect open feedback on the VR experience and the corresponding HMI method.

Finally, the post-test encompassed the same sections of the mid-test questionnaire (to evaluate the second HMI method), together with two additional sections:

- a seventh section that consisted of 10 custom questions where the participants were asked to rank the two tested HMI methods. The ranking aimed to compare the two experiences in terms of naturalness and realism, and focused both on direct interactions (such as opening a zip) and indirect ones (activation of an object that was held in hand).
- An additional, non-numbered section that was aimed to collect open feedback on the two experiences and the two HMI methods.

5 RESULTS AND DISCUSSION

All the data were collected and analyzed using MS Excel. The *t*-test was used to perform comparisons, where not stated otherwise.

Regarding the pre-test results, no significant differences were found between the two groups in terms of age and previous experience with VR technologies; in particular, the participants of the two groups had limited to none previous experience with immersive technologies (on a 1-to-5 scale with Group C: $M = 1.86$ and $D = 0.35$, Group H: $M = 1.86$ and $D = 0.59$, $p = 0.897$).

As for the comparison between the two HMI methods (mid-test and post-test questionnaires), and in particular the first section of the questionnaires, no significant differences were found over the SUS total score and the usability of both experiences was deemed "Good" by the participants (hand-tracking: $M = 81.5$ and $D = 11.17$, controller-based: $M = 84.33$ and $D = 8.87$, $p = 0.375$). Looking at the individual questions, however, it is possible to notice two significant differences in favour of the controller-based experience, that was found less complex w.r.t. the hand-tracking one (hand-tracking: $M = 1.80$ and $D = 0.98$, controller-based: $M = 1.20$ and $D = 0.40$, $p = 0.033$) and led the participants to be more confident (hand-tracking: $M = 3.66$ and $D = 0.69$, controller-based: $M = 4.33$ and $D = 0.59$, $p = 0.003$).

Moving to the second section, that investigated the feeling of Presence, the participants were asked two questions on the coherence of the immersive experience with the real world. In particular, the first question focused on the activation of an object held in hand (e.g., using a liquid dispenser to decontaminate a person), whereas the second question investigated the coherence of direct interactions such as the grabbing and movement of objects in the scene (e.g., opening a zip or removing a filter from the mask). No difference was found for the former, whereas the hand-tracking experience was deemed significantly more coherent to a real-world scenario for direct interactions, since the hand-tracking technology allowed to perform exactly the same gesture required in reality (hand-tracking: $M = 4.53$ and $D = 0.49$, controller-based: $M = 3.80$ and $D = 0.75$, $p = 0.003$). No other difference was found over the remaining statements.

For what it concerns the third and fourth sections, investigating respectively the perceived enjoyment and the attractiveness/hedonic

quality stimulation of the immersive experience (with a focus on the adopted HMI methods), no significant differences were found. In particular, the overall trend of the fourth section, showing that both experiences received positive evaluations over the considered categories, is described in Fig. 4.

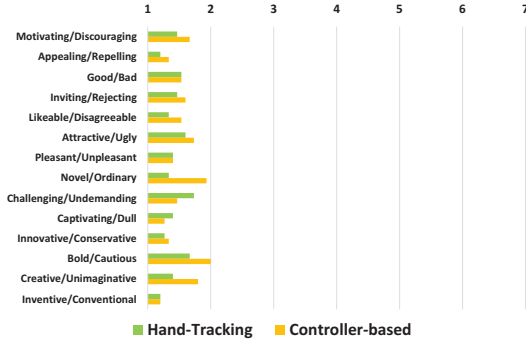


Figure 4: Results of the AttrakDiff questionnaire. For each pair of terms, a low value corresponds to a positive evaluation.

Looking at the results of fifth section, it is possible to highlight some significant differences in favour of the controller-based experience over the hand-tracking one; in particular, the former had a more acceptable response to the user's inputs (hand-tracking: $M = 3.87$ and $D = 0.62$, controller-based: $M = 4.60$ and $D = 0.61$, $p = 0.001$), was found to be less error-prone (hand-tracking: $M = 2.47$ and $D = 1.09$, controller-based: $M = 1.60$ and $D = 0.87$, $p = 0.009$), and was more stable in general (hand-tracking: $M = 1.80$ and $D = 0.75$, controller-based: $M = 1.27$ and $D = 0.44$, $p = 0.027$). However, it is worth noticing that the hand-tracking experience was deemed to have better visual feedback (hand-tracking: $M = 4.87$ and $D = 0.34$, controller-based: $M = 4.60$ and $D = 0.49$, $p = 0.041$) mainly due to the realistic, real-time animations of the hand models, and was considered to be significantly more consistent with real-world experiences (hand-tracking: $M = 4.47$ and $D = 0.62$, controller-based: $M = 4.20$ and $D = 0.65$, $p = 0.041$).

Moving to the sixth section, investigating the mental workload, both the controller-based experience and the hand-tracking one were positively evaluated over every statement (on a 1-to-10 scale, where higher numbers are associated with a higher load). No significant differences were found, except for the perceptual strain (hand-tracking: $M = 1.60$ and $D = 0.87$, controller-based: $M = 2.40$ and $D = 1.25$, $p = 0.028$), probably because the hand-tracking experience offered realistic and natural visual feedback (e.g., the real-time animations of the hands), and the task control (hand-tracking: $M = 2.80$ and $D = 1.47$, controller-based: $M = 1.73$ and $D = 0.67$, $p = 0.005$), probably due to the fact that the controller-based experience was more stable overall.

Finally, looking at the custom questions of the seventh section (Table 1), it is possible to notice several significant differences between the two experiences. On the one hand, the hand-tracking one was deemed more natural when performing direct manipulations such as interacting with objects in the virtual environment (Item 2), opening a zip (Item 5), removing clothes (Item 6) and apply protections (Item 7), and offered a better perception of the hand and the fingers position (Item 3); on the other hand, the controller-based experience was better when the users were asked to activate an object that was held in hand (namely, using a liquid dispenser, Item 4), probably due to the fact that pressing a button on the controller was

similar to the corresponding action in the real world. No differences were found regarding the comfort of the experiences (Item 1), the perceived control during the task (Item 8), the efficiency of the considered HMI method (Item 9), and the preferred HMI method (Item 10).

Item	Hand-Tracking	Controller	<i>p</i> -value
Item 1	$M = 0.47$	$M = 0.53$	0.806
Item 2	$M = 0.87$	$M = 0.13$	0.001
Item 3	$M = 0.73$	$M = 0.27$	0.014
Item 4	$M = 0.13$	$M = 0.87$	0.001
Item 5	$M = 0.93$	$M = 0.07$	< 0.001
Item 6	$M = 1.00$	$M = 0.00$	< 0.001
Item 7	$M = 0.87$	$M = 0.20$	0.003
Item 8	$M = 0.27$	$M = 0.73$	0.068
Item 9	$M = 0.40$	$M = 0.67$	0.301
Item 10	$M = 0.67$	$M = 0.33$	0.207

Table 1: Section seven, mean values and *p*-values obtained using the t-test. Bold font is used to highlight the significant *p*-values ($p < 0.050$). For each item, participants were asked to rank the two experiences; one point was assigned to the first-placed experience, zero points to the second-placed one.

Considering all the results, it emerges that the considered hand-tracking technology is affected by a series of limitations that hinder the overall quality of the experience. In particular, this HMI method is considered significantly more complex than the controllers and is characterized by a less acceptable response to the user's inputs. This leads the users to make more mistakes, resulting in an experience that is perceived to be less controllable w.r.t. the controller based-one (e.g., sometimes it is difficult to start an interaction). However, these limitations are caused mainly by the relative novelty of the hand-tracking system of the Oculus Quest 2 and are merely technological; for example, the HMD uses integrated cameras for the tracking, and it fails to detect the hands when they are at the limits of the cameras field of view or when the hands are partially obstructed. Moreover, the aforementioned problems affect the experience only in a marginal way, since the overall usability of the two versions of the application is not significantly different.

Putting aside these technological problems, it is possible to notice a series of advantages associated with the use of the hand-tracking over the controllers: using the hands to interact directly with the virtual environment, in fact, results in a more natural experience that is more consistent with the real world, confirming what is stated in [1]. When asked to rank the two experiences, in fact, users preferred the hand-tracking over the controllers for all the interaction that required to directly manipulate a virtual object, whereas they preferred the controllers only for the activation of an object held in hand (e.g., using a liquid dispenser), since holding the controller and pressing a button was very similar to the real action. This result was coherent with the trend seen in [12], where users expressed the need for more natural interactions. Furthermore, seeing hand models that are animated in real time and mimic the real hands leads the user to experience less perceptual strain. The above considerations are further confirmed by the open feedback, where participants praised the hand-tracking experience for its immersion and realism, for the immediacy of the control system, and for the fact that, apparently, using the hands to interact directly with the application led to a limited motion sickness w.r.t. the use of the controllers.

6 CONCLUSIONS

This paper presented an investigation on affordable manipulation in VR simulations, focusing on a comparison between hand-tracking and controller-based interactions. The targeted hardware is the

Oculus Quest 2, an affordable HMD that offers both types of interactions and can be considered, nowadays, the go-to for developing immersive VR applications. The study leveraged an immersive VR simulation for training CBRN operators in the CCA procedure; this procedure was chosen due to the variety of manual actions that the user must perform to successfully complete the assigned tasks. The application was used in two versions, identical except for the interaction logic.

A within-subject user study was conducted involving 15 participants, and subjective measures were collected leveraging state-of-the-art questionnaires together with custom questions. Based on the results, it emerged that the hand tracking technology of the Oculus Quest 2 suffers from a series of technological limitations that led to some inconsistencies and errors, without hindering the overall usability of the system. Moreover, it emerged that the use of the hand-tracking technology resulted in a more natural experience that was more consistent with the real-world w.r.t. the controller-based version, in particular when the participants were requested to interact directly and manipulate virtual objects, leading to a more pronounced immersion.

The current study is characterized by some limitations, such as the fact that the simulation was only adapted to be used in the study, but was not designed for it. In the future, it would be interesting to devise an application for the sole purpose of testing different types of interactions to extend the study and consider other types of manipulation, e.g., including virtual objects of different size. Another limitation is the fact that only the Oculus Quest 2 was considered for this study; it would be useful to include other, less-known devices or hand-tracking technologies [17] to offer a wider and more in-depth analysis. Finally, as a new research direction, it could be worth to investigate if different HMI methods have different impact on the educational aspects of a VR training application, in particular to understand if more natural interactions could improve the learning gain of the trainees.

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REFERENCES

- [1] E. Bouzbib, G. Bailly, S. Haliyo, and P. Frey. "can I touch this?": Survey of virtual reality interactions via haptic solutions. *CoRR*, abs/2101.11278, 2021.
- [2] J. Brooke. SUS: A 'quick and dirty' usability scale. *Usability Evaluation in Industry*, p. 189, 1996. doi: 10.1201/9781498710411-35
- [3] F. Buttussi and L. Chittaro. A comparison of procedural safety training in three conditions: Virtual reality headset, smartphone, and printed materials. *IEEE Transactions on Learning Technologies*, 14(1):1–15, 2021. doi: 10.1109/TLT.2020.3033766
- [4] F. De Lorenzis, F. G. Praticò, M. Repetto, E. Pons, and F. Lamberti. Immersive virtual reality for procedural training: Comparing traditional and learning by teaching approaches. *Computers in Industry*, 144:103785, 2023.
- [5] M. Edwards, N. Baloian, and J. A. Pino. Hand gesture preferences analysis for a virtual reality scenario. *Lecture Notes in Networks and Systems*, 594 LNNS:900 – 912, 2023.
- [6] P. Fratzak, Y. M. Goh, P. Kinnell, L. Justham, and A. Soltoggio. Virtual reality study of human adaptability in industrial human-robot collaboration. pp. 1–6, 2020. doi: 10.1109/ICHMS49158.2020.9209558
- [7] D. Hamilton, J. McKechnie, E. Edgerton, and C. Wilson. Immersive virtual reality as a pedagogical tool in education: a systematic literature review of quantitative learning outcomes and experimental design. *Journal of Computers in Education*, 8(1):1–32, 2021.
- [8] D. Harris, M. Wilson, and S. J. Vine. Development and validation of a simulation workload measure: The simulation task load index (sim-tlx). Dec 2019. doi: 10.31234/osf.io/pm5ht
- [9] M. Hassenzahl, F. Koller, and M. Burmester. Der user experience (UX) auf der spur: Zum einsatz von www.attrakdiff.de. In H. Brau, S. Diefenbach, M. Hassenzahl, F. Koller, M. Peissner, and K. Röse, eds., *Tagungsband UP08*, pp. 78–82. Fraunhofer Verlag, Stuttgart, 2008.
- [10] P. Jost, S. Cobb, and I. Hämmerle. Reality-based interaction affecting mental workload in virtual reality mental arithmetic training. *Behaviour & Information Technology*, 39(10):1062–1078, 2020. doi: 10.1080/0144929X.2019.1641228
- [11] R. S. Kalawsky. VRUSE – A computerised diagnostic tool: For usability evaluation of virtual/synthetic environment systems. *Applied Ergonomics*, 30(1):11–25, 1999. doi: 10.1016/S0003-6870(98)00047-7
- [12] F. Lamberti, F. De Lorenzis, F. G. Praticò, and M. Migliorini. An immersive virtual reality platform for training CBRN operators. In *Proc. of 2021 IEEE 45th Annual Computers, Software, and Applications Conference*, pp. 133–137, 2021. doi: 10.1109/COMPSAC51774.2021.00030
- [13] S. Nilsson, T. Nilsson, L. Russell, B. Costa, E. Pieters, J. Winter, C. Fernandez, and C. Myhre. Fruit golf: An asymmetrical shared space vr/mobile experience. *Proceedings - SIGGRAPH 2022 Immersive Pavilion*, 2022. Cited by: 0. doi: 10.1145/3532834.3536214
- [14] F. G. Praticò, F. De Lorenzis, D. Calandra, A. Cannavò, and F. Lamberti. Exploring simulation-based virtual reality as a mock-up tool to support the design of first responders training. *Applied Sciences*, 11(16):1–13, 2021. doi: 10.3390/app11167527
- [15] A. Sutcliffe, B. Gault, and J.-E. Shin. Presence, memory and interaction in virtual environments. *International Journal of Human-Computer Studies*, 62(3):307–327, 2005.
- [16] S. T. Tokel and V. Isler. Acceptance of virtual worlds as learning space. *Innovations in Education and Teaching International*, 52(3):254–264, 2015.
- [17] J. Wang, F. Müller, F. Bernard, S. Sorli, O. Sotnychenko, N. Qian, M. A. Otaduy, D. Casas, and C. Theobalt. Rgb2hands: Real-time tracking of 3d hand interactions from monocular RGB video. *CoRR*, abs/2106.11725, 2021.
- [18] C. Wee, K. M. Yap, and W. N. Lim. Haptic interfaces for virtual reality: Challenges and research directions. *IEEE Access*, 9:112145–112162, 2021. doi: 10.1109/ACCESS.2021.3103598
- [19] S.-C. Yeh, E. H.-K. Wu, Y.-R. Lee, R. Vaitheeshwari, and C.-W. Chang. User experience of virtual-reality interactive interfaces: A comparison between hand gesture recognition and joystick control for xrspace manova. *Applied Sciences (Switzerland)*, 12(23), 2022. doi: 10.3390/app122312230